

[RI/2024/Q3]

3 Name your Jupyter Notebook as:

TASK3_<your name>_<class>_<index number>.ipynb

Implement a college management system using Object-Oriented Programming.

The system will manage two types of entities: `Student` and `Staff`. These entities will be created as subclasses which inherit from the base class `Person`.

Both entities will be stored and managed in two separate linked list structures: `StudentLinkedList` and `StaffLinkedList`. These linked lists will inherit from the base class `LinkedList`.

The class `Person` contains the following attributes:

- `name`: the name of the person.
- `personID`: the unique ID of the person.

The class `Person` contains a method `getDetails()` that returns a string containing the person's name and ID.

The `Student` class inherits from the `Person` class. Add the following additional attributes to the `Student` class:

- `course`: the course the student is enrolled in.
- `year`: the year of study.

The `Student` class contains a method `getDetails()` that returns a string containing the student's name, ID, course and year of study.

The `Staff` class inherits from the `Person` class. Add the following additional attributes to the `Staff` class:

- `department`: the department the staff member belongs to.
- `position`: the position or job title of the staff member.
- `salary`: the salary of the staff member.

The `Staff` class contains a method `getDetails()` that returns a string containing the staff member's name, ID, department, position and salary.

For the sub-task, add a comment statement at the beginning of the code using the hash symbol '#', to indicate the sub-task the program code belongs to, for example:

```
In [1]: #Task 3.1
        Program code
```

Output:

Task 3.1

Write program code to declare the classes `Person`, `Student` and `Staff`, with all their attributes and methods.

[7]

[RI/2024/Q3 Cont ']

Task 3.2

The class `LinkedList` contains the following attributes:

- `Data`: the data stored in the node, which can be student or staff
- `Pointer`: points to the next node in the linked list

The class `LinkedList` contains the following methods:

- a constructor to set the `Data` to its parameter, and pointer to `None`.
- `insert_rec(data)`: recursive method to insert a new node with the given data at the end of the linked list, where the parameter represents any object.
- `delete_rec(data)`: recursive method to delete the node with the specified data from the linked list, where the parameter identifies the node to be deleted.
- `display_rec()`: recursive method to display the details of all nodes in the linked list.

Write program code to declare the class `LinkedList` with all its attributes and methods. [11]

Task 3.3

The `StudentLinkedList` class inherits from the `LinkedList` class and contains the following additional attributes:

- `studentCount` that keeps track of the total number of students in the linked list.
- `courseFreq` that keeps track of the number of students in each course.

The `StudentLinkedList` class contains the following methods:

- `updateCourseFreq` that will update the `courseFreq`.
- `displayCourseFreq` that will display the number of students from each course.
- `insert(data)` method to insert a new node data at the end of the linked list, where the parameter is a `Student` object.
- `delete(data)` method to delete a specified node data from the linked list, where the parameter identifies the node to be deleted.

The `studentCount` and `courseFreq` will be updated whenever a node data is inserted or deleted.

[RI/2024/Q3 Cont ']

The `StaffLinkedList` class inherits from the `LinkedList` class and contains the following additional attribute:

- `staffCount` that keeps track of the number of staff members.

The `StaffLinkedList` class contains the following methods:

- `insert(data)` method to insert a new node data at the end of the linked list, where the parameter is a `Staff` object.
- `delete(data)` method to delete a specified node data from the linked list, where the parameter identifies the node to be deleted.

The `staffCount` will be updated whenever a node data is inserted or deleted.

Write program code to declare the classes `StudentLinkedList` and `StaffLinkedList`, with all their attributes and methods. [11]

Task 3.4

Write and execute code to test your program according to the instructions specified in the text file “TASK3_TESTING.txt”. [2]

Save your Jupyter Notebook for Task 3.